

## CRAIG-MOFFAT, CO, USA

WMO: 725700

Lat: 40.493N

Lon: 107.524W

Elev: 1887

StdP: 80.62

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24046

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -25.3      | -22.3 | -29.0                         | 0.3 | -24.4 | -25.6 | 0.5 | -21.9 | 10.9                  | 2.4  | 9.2 | 0.9  | 0.4                   | 280  | 0.645 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.0                   | 32.9            | 14.3 | 31.7 | 14.1 | 30.2 | 13.8 | 16.5                | 27.0 | 15.8 | 26.4 | 15.2 | 25.9 | 3.8                  | 260  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 13.6                            | 12.3 | 18.0 | 12.6 | 11.4 | 18.1 | 11.4 | 10.6 | 18.2 | 52.9          | 26.7 | 50.8 | 26.4 | 48.7 | 25.7 | 19.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 10.9              | 9.0  | 7.8 | -31.2                      | 35.4 | 4.2                | 1.2 | -34.2                                              | 36.2 | -36.7      | 36.9 | -39.0      | 37.6 | -42.0      | 38.4 |    |    |
|                   |      |     | -31.0                      | 17.7 | 3.5                | 0.9 | -33.5                                              | 18.4 | -35.6      | 18.9 | -37.6      | 19.5 | -40.1      | 20.1 |    |    |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 6.0    | -8.6  | -5.3  | 0.9   | 5.8   | 10.6  | 15.9  | 20.1  | 18.7  | 13.8  | 6.6   | -0.2  | -7.4  |
|                                                                   | DBStd       | 10.68  | 6.25  | 5.85  | 5.01  | 3.87  | 3.67  | 3.23  | 2.22  | 2.11  | 3.50  | 4.40  | 5.50  | 6.08  |
|                                                                   | HDD10.0     | 2430   | 577   | 428   | 282   | 132   | 38    | 3     | 0     | 0     | 9     | 118   | 306   | 538   |
|                                                                   | HDD18.3     | 4622   | 835   | 661   | 541   | 377   | 239   | 85    | 9     | 20    | 140   | 363   | 556   | 796   |
|                                                                   | CDD10.0     | 959    | 0     | 0     | 1     | 5     | 57    | 178   | 312   | 271   | 123   | 13    | 0     | 0     |
|                                                                   | CDD18.3     | 109    | 0     | 0     | 0     | 0     | 1     | 11    | 62    | 33    | 3     | 0     | 0     | 0     |
|                                                                   | CDH23.3     | 3663   | 0     | 0     | 0     | 2     | 87    | 720   | 1482  | 1024  | 334   | 13    | 0     | 0     |
|                                                                   | CDH26.7     | 1433   | 0     | 0     | 0     | 0     | 14    | 258   | 683   | 393   | 84    | 1     | 0     | 0     |
| Wind                                                              | WSAvg       | 2.4    | 1.6   | 2.1   | 2.6   | 3.2   | 2.9   | 2.9   | 2.5   | 2.4   | 2.3   | 2.3   | 2.0   | 1.7   |
| Precipitation                                                     | PrecAvg     | 418    | 29    | 28    | 31    | 46    | 46    | 27    | 30    | 33    | 44    | 41    | 34    | 29    |
|                                                                   | PrecMax     | 546    | 63    | 56    | 53    | 94    | 135   | 90    | 69    | 100   | 104   | 84    | 73    | 87    |
|                                                                   | PrecMin     | 274    | 1     | 5     | 1     | 20    | 7     | 6     | 0     | 5     | 13    | 10    | 12    | 10    |
|                                                                   | PrecStd     | 78     | 15    | 12    | 14    | 19    | 29    | 21    | 16    | 18    | 24    | 18    | 14    | 17    |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 7.1   | 10.2  | 20.2  | 22.8  | 28.4  | 33.8  | 35.0  | 33.5  | 31.4  | 25.1  | 18.4  | 10.4  |
|                                                                   |             | MCWB   | 1.9   | 3.6   | 7.4   | 9.0   | 12.4  | 14.0  | 15.0  | 14.6  | 14.1  | 10.6  | 6.8   | 3.6   |
|                                                                   | 2%          | DB     | 4.1   | 7.1   | 16.4  | 20.9  | 25.8  | 31.6  | 33.5  | 32.1  | 28.9  | 22.5  | 15.2  | 6.1   |
|                                                                   |             | MCWB   | 0.6   | 2.2   | 6.0   | 8.0   | 11.3  | 13.3  | 14.6  | 14.1  | 12.6  | 9.6   | 5.6   | 1.7   |
|                                                                   | 5%          | DB     | 2.5   | 5.0   | 13.3  | 18.7  | 23.6  | 29.7  | 32.3  | 30.8  | 27.3  | 20.2  | 12.3  | 3.6   |
|                                                                   |             | MCWB   | -0.1  | 1.2   | 4.8   | 7.1   | 10.5  | 12.8  | 14.5  | 13.9  | 12.1  | 8.7   | 4.6   | 0.3   |
|                                                                   | 10%         | DB     | 1.1   | 3.0   | 10.2  | 16.3  | 21.5  | 27.7  | 30.9  | 29.0  | 25.1  | 17.4  | 9.1   | 1.6   |
|                                                                   |             | MCWB   | -0.7  | 0.0   | 3.6   | 6.3   | 9.9   | 12.6  | 14.4  | 13.6  | 11.5  | 7.6   | 3.2   | -0.9  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 2.7   | 4.1   | 7.9   | 10.0  | 14.1  | 15.7  | 17.5  | 17.3  | 15.6  | 11.9  | 7.6   | 4.0   |
|                                                                   |             | MCDB   | 5.3   | 9.1   | 18.7  | 21.4  | 25.6  | 27.5  | 27.6  | 27.5  | 26.6  | 21.3  | 16.1  | 9.2   |
|                                                                   | 2%          | WB     | 1.3   | 2.6   | 6.5   | 8.6   | 12.3  | 14.8  | 16.7  | 16.3  | 14.4  | 10.6  | 6.1   | 2.1   |
|                                                                   |             | MCDB   | 3.4   | 6.3   | 15.5  | 18.6  | 22.7  | 27.3  | 27.4  | 26.1  | 24.1  | 20.1  | 14.3  | 5.5   |
|                                                                   | 5%          | WB     | 0.3   | 1.5   | 5.1   | 7.6   | 11.3  | 14.0  | 16.1  | 15.7  | 13.5  | 9.5   | 5.0   | 0.6   |
|                                                                   |             | MCDB   | 2.2   | 4.6   | 12.6  | 17.3  | 21.5  | 26.3  | 27.1  | 25.6  | 22.9  | 18.5  | 11.8  | 3.1   |
|                                                                   | 10%         | WB     | -1.0  | 0.4   | 3.8   | 6.6   | 10.4  | 13.2  | 15.5  | 15.0  | 12.6  | 8.2   | 3.5   | -0.7  |
|                                                                   |             | MCDB   | 1.0   | 2.9   | 9.8   | 15.3  | 20.0  | 25.2  | 26.4  | 25.0  | 21.8  | 16.3  | 8.5   | 1.4   |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 13.5  | 13.2  | 14.4  | 15.6  | 17.4  | 21.0  | 21.0  | 20.3  | 20.2  | 17.9  | 15.1  | 13.2  |
|                                                                   |             | MCDBR  | 13.1  | 13.5  | 19.5  | 22.1  | 22.7  | 25.2  | 24.1  | 24.0  | 24.7  | 23.6  | 20.3  | 13.7  |
|                                                                   |             | MCWBR  | 10.0  | 9.3   | 10.3  | 10.7  | 10.0  | 9.9   | 8.4   | 8.7   | 10.2  | 12.0  | 11.9  | 9.7   |
|                                                                   | 5% WB       | MCDBR  | 11.6  | 12.2  | 18.0  | 19.7  | 20.2  | 21.6  | 19.1  | 18.3  | 19.0  | 20.8  | 18.9  | 12.5  |
|                                                                   |             | MCWBR  | 9.0   | 8.6   | 9.8   | 9.9   | 9.1   | 8.9   | 6.9   | 7.0   | 8.2   | 10.8  | 11.2  | 9.0   |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.235 | 0.248 | 0.270 | 0.304 | 0.312 | 0.316 | 0.346 | 0.345 | 0.297 | 0.271 | 0.250 | 0.240 |
|                                                                   | taud        |        | 2.497 | 2.508 | 2.532 | 2.453 | 2.493 | 2.474 | 2.439 | 2.433 | 2.570 | 2.617 | 2.584 | 2.495 |
|                                                                   | Ebn at Noon |        | 957   | 989   | 995   | 971   | 964   | 955   | 923   | 916   | 950   | 950   | 933   | 921   |
|                                                                   | Edh at Noon |        | 72    | 84    | 93    | 108   | 106   | 109   | 112   | 109   | 89    | 75    | 67    | 67    |
| All-Sky Solar Radiation                                           | RadAvg      |        | 2.21  | 3.02  | 4.38  | 5.43  | 6.38  | 7.67  | 7.16  | 6.21  | 5.24  | 3.65  | 2.42  | 1.83  |
|                                                                   | RadStd      |        | 0.15  | 0.22  | 0.42  | 0.37  | 0.47  | 0.45  | 0.30  | 0.27  | 0.33  | 0.30  | 0.27  | 0.16  |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (16 neighbors) |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page